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Backshoring of production in the context of a small and open Nordic economy

Small and open
Nordic
economy

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Abstract

Purpose – The purpose of this paper is to investigate the extent, drivers, and conditions underlying backshoring in the Finnish manufacturing industry, comparing the results to the wider ongoing relocation of production in the international context.

Design/methodology/approach – The survey of 229 Finnish manufacturing firms reveals the background, drivers, and patterns of offshoring and backshoring.

Findings – Companies that had transferred their production back to Finland were more commonly in industries with relatively higher technology intensity and they were typically larger than the no-movement companies, and with a higher number of plants. They also reported more commonly having a corporate-wide strategy for guiding production location decisions.

Research limitations/implications – Backshoring activity in the small and open economy of Finland seems to be higher compared to earlier studies in larger countries. The findings suggest that there is a transformation in the manufacturing industries with some gradual replacement of labor-intensive and lower technology-intensive industries toward higher technology-intensive industries.

Practical implications – Moving production across national borders is one option in the strategies of firms to stay competitive. Companies must carefully consider the relevance of various decision-making drivers when determining strategies for their production networks.

Social implications – Manufacturing industries have traditionally been important for employment in the relatively small and open economies of the Nordic countries. From the social perspective, it is important to understand the ongoing transformation and its implications.

Originality/value – There are few empirical studies available of the ongoing backshoring movement, utilizing data from company decision makers instead of macroeconomic factors.

Keywords Production, Manufacturing, Manufacturing strategy

Paper type Research paper

1. Introduction

Manufacturing industries have a central role in economic growth through production and its interplay with wider business ecosystems. Therefore, it is necessary to understand the mechanisms leading to manufacturing companies' production location decisions. This research paper describes the case of the Finnish manufacturing industry and the extent to which Finnish-based manufacturing companies move their production outside of Finland (offshoring) and bring their production back (backshoring).

The goal of this research is to enhance knowledge of the drivers and characteristics of backshoring, and its implications on the renewal of manufacturing industry in a small and open economy, such as Finland. The focus is on two main research questions:

RQ1. Why and to what extent do Finnish manufacturing firms offshore and backshore their production?

This research was conducted in collaboration with Professors Jan Olhager from the Lund University in Sweden and Jan Stentoft from the University of Southern Denmark. The research was part of the Innovation research program "Renewal of Manufacturing" jointly financed by Tekes – the Finnish Funding Agency for Innovation and the Swedish innovation agency VINNOVA. The authors acknowledge the financial support for this research, and the Nordic research partners for the fruitful cooperation.



RQ2. How do the backshoring companies differ from other companies?

The research was carried out as part of a three-country survey involving Denmark, Finland, and Sweden, of the trends and drivers of offshoring and backshoring of manufacturing. The particular focus in this paper is on the companies that have brought their production back to Finland during the period 2010-2015.

Extant literature has reported numerous reasons for moving manufacturing back to the developed economies from a previous location (Stentoft *et al.*, 2016). Different drivers have been identified for backshoring, such as changing costs (Bailey and De Propriis, 2014; Cho *et al.*, 2014; Kinkel, 2012, 2014; Pearce, 2014; Martinez-Mora and Merino, 2014; Wu and Zhang, 2014), quality issues (Canham and Hamilton, 2013; Fratocchi *et al.*, 2014, 2016; Gylling *et al.*, 2015; Wiesmann *et al.*, 2017), and time and flexibility issues (Arlbjørn and Mikkelsen, 2014; Fratocchi *et al.*, 2014, 2016; Kinkel and Maloca, 2009; Stentoft *et al.*, 2015). There are only a few studies available in which the backshoring phenomenon has been examined in the context of small countries (cf. Canham and Hamilton, 2013; Stentoft *et al.*, 2015). Future research is recommended on backshoring to explore differences in backshoring practices with respect to industry comparisons, firm-size and country practices based on empirical data (Barbieri and Stentoft, 2016).

In this paper, we first review the related literature. Then, we present the research methodology and means of collecting data. The main section of the paper is concerned with the scope and drivers of offshoring and backshoring of the Finnish manufacturing industry and comparison of companies that have brought production back to Finland with those that have not done so. We conclude with implications for research and practice as well as discuss limitations and suggestions for further research.

2. Literature review

2.1 Manufacturing industry in the Nordic countries

Manufacturing firms' competitiveness requires sufficiently concentrated clusters of resources – interconnected manufacturing firms, suppliers, service providers, and associated institutions that collaborate (Porter, 2000). The understanding used to be that proximity in geographic and various other institutional terms is good for accessing resources, building relationships, and achieving productivity and related benefits (Porter, 2000). Therefore, large countries with easy access domestically to markets, raw materials, and human resources, innovation capacity and economies of scale have dominated in the manufacturing landscape. However, globalization and internationalization of firms have changed this constellation, as the costs of transportation and information exchange have diminished dramatically, resulting in a changing configuration of production resources (Porter, 2000; Brennan *et al.*, 2015). Resources and markets may thus be globally available to any firms, and at reasonable costs. Trends such as servitization, creation of new production systems, and sustainability and ethicality of the supply chains are driving changes in manufacturing networks, thereby potentially changing the competitive arena and generating new motives for the relocation of manufacturing (Brennan *et al.*, 2015).

A lot of attention in research on the internationalization of manufacturing has been on large developed countries (e.g. Ellram *et al.*, 2013; Gray *et al.*, 2011; Kinkel and Maloca, 2009; Martinez-Mora and Merino, 2014; Pearce, 2014; Porter and Rivkin, 2012). This paper adopts the perspective of a small, developed country, specifically in the Nordic region. Such context has been covered only to a limited extent in previous relocation research (e.g. Arlbjørn and Mikkelsen, 2014; Gylling *et al.*, 2015; Stentoft *et al.*, 2015). In Nordic countries, the main markets and demand for manufactured goods are often located at a distance. Labor costs tend to be relatively high, labor rules and laws are relatively inflexible, and these factors have encouraged manufacturing firms to establish new capacity in the emerging markets,

where also the labor costs are lower. This has resulted in job losses in the home countries (Alsén *et al.*, 2013). With high-technology manufacturing, also the access to specialized raw materials and components, and manual labor has required locational considerations (Groth and Winther, 2013).

These issues have driven manufacturing firms and industries to consider their international presence actively. Although Nordic countries (Denmark, Finland, Norway, Sweden) have been considered as active in their cluster-based economic development initiatives (Porter, 2000) and they are praised for their access to talented workforce (Alsén *et al.*, 2013), the economic success of the small Nordic countries is heavily dependent on exports (Alsén *et al.*, 2013). Therefore, manufacturing industries have been an important source of economic wealth. The decline in the manufacturing sector has caused significant concerns for the welfare of these economies.

The share of the manufacturing sector in the Nordic countries' gross national product (GNP) has declined in recent decades, following the same trend as in other developed economies. In Finland, for example, the highest manufacturing share of GNP was reached in 1974, when it accounted for 32 percent of GNP. In the early 2000s the share fluctuated between 25 and 28 percent, but fell to 20 percent in 2009 as a result of the global financial crisis that started in 2008. The share has remained the same since then, and the other sectors have not been able to compensate for the loss of this contribution to GNP (Finland's Ministry of Employment and the Economy (FMEE), 2014).

Employment in manufacturing industries has also been in decline. In 2007, the manufacturing sector in Finland employed directly 397,000 people. The reduction in the manufacturing workforce between 2007 and 2016 was 90,000 people, i.e. 23 percent, resulting in 307,000 people directly employed in the manufacturing industry in 2016. Over the long term, however, Nordic manufacturing output has continued to grow, primarily because of investments in automation and the shift toward more technology-intensive and knowledge-intensive sectors.

Offshoring and job losses have mostly affected the labor-intensive industries, such as furniture, textiles, and electrical components. Much of production in these industries has moved to lower cost countries and emerging markets. For example, Swedish computing and electrical component manufacturers employed virtually all their workers domestically in 2000. A decade later, 22 percent of their employees were in low-cost countries. In addition, a large number of workers were employed offshore by contract manufacturers. In the textile and clothing industries, the share of employment in low-cost countries rose from 8 to 42 percent. The labor force of Swedish machinery and equipment companies in low-cost countries rose from 15 percent in 1998-2000 to 34 percent in 2008-2010 (Alsén *et al.*, 2013).

In the changing global business environment, the Nordic manufacturing companies are increasingly shifting the focus of their investments outside their home bases. From 2000 to 2006, 81 percent of capital expenditure by Finnish manufacturing companies was invested within the Nordic region. From 2007 to 2011 that dropped to 67 percent. Finnish capital investments have been moving to the rest of Western Europe and to the developing economies, including Russia, Eastern Europe, Asia, and South America (FMEE, 2014).

The decreased industrial value-added is the most significant challenge that the manufacturing sector in Finland is presently facing. The Finnish electronics industry lost about EUR9 billion of its value-added during the period 2009-2015. In the wood and paper industry the loss has been about EUR1.5 billion and the metal processing industry has also been severely affected. The causes are slightly different across the three industries. In the electronics industry the loss of value-added could be explained by the collapse and eventual divestment of Nokia's mobile terminals business to Microsoft. The loss in the paper industry was caused by the global decrease in the demand for printed media, with

the resulting overcapacity and the need to close factories in the Nordic countries. In metal processing there was a combination of factors behind the loss of industrial value-added: lower demand for investment goods, increased competition from the emerging markets, lowered prices for metal products, and loss of competitiveness because of increasing cost levels (FMEE, 2014). With these background factors, the dependency of Nordic economies on manufacturing competitiveness makes it necessary to understand the premises for successful backshoring.

2.2 Offshoring and backshoring of manufacturing

Much research has been published on offshoring and outsourcing. Companies choose these actions for several reasons, for example, to obtain cost advantages and proximity to customers (Kakabadse and Kakabadse, 2002; Kinkel and Maloca, 2009). More recently research has focused on the reverse movement, that is, moving manufacturing back to its original location, known as backshoring (Kinkel, 2014), reshoring (Gray *et al.*, 2013; Ellram *et al.*, 2013; Tate, 2014), or re-insourcing (Drauz, 2014).

Stentoft *et al.* (2016) made a content-based literature review analyzing the research so far on the backshoring of manufacturing. The specific reasons for moving manufacturing back to manufacturing companies' home countries can be related to six distinct aspects: cost; quality; time and flexibility; access to skills and knowledge; risks; and other factors (such as incentives, core focus, shrinking market, and rectification of an erroneous decision), see Table I for literature references.

Backshoring operations have been studied in various country contexts, and the target and source countries have been considered. Kinkel (2012) conducted a survey with almost 1,500 German firms among various industries and firm sizes. Besides covering the motives of offshoring and backshoring and changes in these after the economic crisis, the findings showed that the German manufacturing firms had most frequently offshored to Eastern European countries, China, and other Asian countries, and that this activity has changed (less to Eastern Europe and more to Asia) after the economic crisis. Backshoring has similarly been most active from Eastern Europe and China. Kinkel also reported a decline in offshoring activities after the economic crisis of 2008.

There have been some backshoring studies focusing on certain types of businesses, or certain countries only. For example, Zhai *et al.* (2016) focused on manufacturing firms that had backshored from China to the USA, using a document-based study of 139 backshoring cases. They covered a variety of industries. Robinson and Hsieh (2016) used a qualitative interview-based study and focused on the manufacturing of luxury clothing, specifically Burberry and its supply partners, and the backshoring of manufacturing to the UK. Gylling *et al.* (2015) report an illustrative single case from Finland on first the offshoring and eventually the backshoring of bicycle manufacturing. Bals *et al.* (2016) studied reshoring decision making using documentary data from Germany and the USA, and propose various new research avenues, for example, those concerning the varying home country conditions and how these give rise to differentiation in backshoring patterns and decision-making.

In the context of small countries, i.e. Denmark, Finland, and New Zealand, the following factors favoring backshoring are specifically addressed in the existing research:

- quality not at an acceptable level in the source country (Arlbjørn and Mikkelsen, 2014; Canham and Hamilton, 2013; Gylling *et al.*, 2015; Stentoft *et al.*, 2015);
- delivery lead time (Arlbjørn and Mikkelsen, 2014; Stentoft *et al.*, 2015), demand volatility and supply chain resilience (Gylling *et al.*, 2015; Stentoft *et al.*, 2015);
- proximity to R&D resources, and utilization of new technologies and automation (Arlbjørn and Mikkelsen, 2014; Stentoft *et al.*, 2015);

			Small and open Nordic economy
Drivers	Specific issues	References	
Changing costs of operations	Increasing labor costs	Bailey and De Propriis (2014), Cho <i>et al.</i> (2014), Kinkel (2012, 2014), Pearce (2014), Tate (2014), Tate <i>et al.</i> (2014)	
	Increasing logistics costs	Bailey and De Propriis (2014), Kinkel (2014), Moser (2013), Pearce (2014), Tate (2014)	
	Eroding cost advantage	Bailey and De Propriis (2014), Fratocchi <i>et al.</i> (2014, 2016), Martinez-Mora and Merino (2014), Wu and Zhang (2014)	
	Higher-than-expected coordination efforts and transaction costs	Kinkel (2012, 2014), Kinkel and Maloca (2009), Gray <i>et al.</i> (2013); Gylling <i>et al.</i> (2015)	
	Miscalculation of actual cost	Gylling <i>et al.</i> (2015); Kinkel (2014), Kinkel and Maloca (2009), Martinez-Mora and Merino (2014), Tate <i>et al.</i> (2014)	
	Changes in energy costs	Pearce (2014), Tate (2014), Tate <i>et al.</i> (2014)	
Quality	Productivity differences between locations	Pearce (2014)	
	Need for small production runs	Canham and Hamilton (2013), Gylling <i>et al.</i> (2015)	
	Quality differences between locations	Arlbjørn and Mikkelsen (2014), Bailey and De Propriis (2014), Canham and Hamilton (2013), Fratocchi <i>et al.</i> (2014, 2016); Gylling <i>et al.</i> (2015), Kinkel (2012, 2014), Kinkel and Maloca (2009), Moser (2013), Stentoft <i>et al.</i> (2015), Tate <i>et al.</i> (2014), Zhai <i>et al.</i> (2016), Wiesmann <i>et al.</i> (2017)	
	Customer perceived quality	Canham and Hamilton (2013), Fratocchi <i>et al.</i> (2016)	
	Product quality	Canham and Hamilton (2013), Fratocchi <i>et al.</i> (2016), Stentoft <i>et al.</i> (2015), Tate <i>et al.</i> (2014), Zhai <i>et al.</i> (2016)	
	Process quality	Kinkel and Maloca (2009)	
Time and flexibility	Quality control and management processes	Bailey and De Propriis (2014), Canham and Hamilton (2013), Martinez-Mora and Merino (2014), Kinkel (2012, 2014)	
	Co-location between manufacturing and R&D, innovation, services or development of system solutions to maximize quality	Bailey and De Propriis (2014), Fratocchi <i>et al.</i> (2016)	
	Shorter lead-times	Arlbjørn and Mikkelsen (2014), Bailey and De Propriis (2014), Gylling <i>et al.</i> (2015), Kinkel and Maloca (2009), Martinez-Mora and Merino (2014), Stentoft <i>et al.</i> (2015), Wu and Zhang (2014)	
	On-time delivery	Bailey and De Propriis (2014), Fratocchi <i>et al.</i> (2014), Kinkel (2012)	
	Greater flexibility	Bailey and De Propriis (2014), Fratocchi <i>et al.</i> (2014, 2016), Gylling <i>et al.</i> (2015), Kinkel (2012, 2014), Martinez-Mora and Merino (2014), Stentoft <i>et al.</i> (2015)	
	Co-location between marketing and production for good time-to-customer	Fratocchi <i>et al.</i> (2016)	
Access to skills and knowledge	Proximity to R&D resources	Arlbjørn and Mikkelsen (2014), Bailey and De Propriis (2014), Kinkel (2014), Pearce (2014), Tate (2014), Stentoft <i>et al.</i> (2015)	
	Availability of skilled labor	Bailey and De Propriis (2014), Fratocchi <i>et al.</i> (2016), Kinkel (2012, 2014), Kinkel and Maloca (2009), Pearce (2014), Tate <i>et al.</i> (2014)	
	Utilization of new technologies and automation	Arlbjørn and Mikkelsen (2014), Stentoft <i>et al.</i> (2015), Tate <i>et al.</i> (2014)	
Other specific drivers	The risk of losing know-how or intellectual property	Kinkel (2014), Moser (2013), Tate (2014), Tate <i>et al.</i> (2014)	

Table I.
Summary of previous
research on the
(continued) drivers of backshoring

Table I.

Drivers	Specific issues	References
	Supply chain risks	Gray <i>et al.</i> (2013), Moser (2013), Tate (2014), Ashby (2016), Wiesmann <i>et al.</i> (2017)
	Risks for the customer perception of the brand value	Robinson and Hsieh (2016), Wiesmann <i>et al.</i> (2017)
	Volatility in currency exchange rates	Bailey and De Propriis (2014), Gylling <i>et al.</i> (2015), Moser (2013), Tate <i>et al.</i> (2014)
	Government incentives favoring a certain location	Moser (2013), Pearce (2014), Tate <i>et al.</i> (2014)

- volatility in the currency exchange rates (Gylling *et al.*, 2015), market-related factors, such as loyalty and/or patriotism, value of “Made in X” and staying close to customers (Canham and Hamilton, 2013); and
- increased focus on core activities at the focal manufacturing firm (Arlbjørn and Mikkelsen, 2014; Canham and Hamilton, 2013; Stentoft *et al.*, 2015).

There is some indication that company size is relevant in the decisions to both offshore and backshore manufacturing. Earlier research shows that bigger companies are more active in offshoring than small or medium sized firms, but there is no difference in backshoring (Kinkel, 2012). Srai and Ane (2016) conducted a qualitative survey of 94 companies that had backshored manufacturing to France and the UK. They did not specifically address the impact of size, but their sample included a fairly large proportion of medium-sized and small firms, indicating that SMEs may also be active in backshoring. A broad range of industries have been studied when analyzing offshoring and backshoring (Brennan *et al.*, 2015; Kinkel, 2012; Massini *et al.*, 2010).

There are only a few systematic comparative studies available of the extent and drivers of offshoring and backshoring trends in small and open, export-dependent economies. Also, there are only few studies available in which the research data come from the level of company decision makers; the trends in manufacturing employment and investments have rather been observed through macroeconomic numbers. Simultaneous analysis of offshoring and backshoring using the same set of survey questions has also not been done before. This study seeks to contribute by addressing these gaps with empirical data collected among firms in the Finnish manufacturing industry.

3. Research design and approach

3.1 Design of the questionnaire survey

This paper is based on survey research carried out in Fall 2015 on the extent and characteristics of manufacturing companies’ production offshoring and backshoring activities in Denmark, Finland, and Sweden. The survey design is based on the earlier research on relocation of manufacturing in Germany, the USA, Spain, and Denmark; cf. Kinkel (2014), Tate (2014), Martínez-Mora and Merino (2014), and Stentoft *et al.* (2015). The questionnaire consists of five sections focusing on: first, a company’s manufacturing network characteristics; second and third, the extent and perceived drivers, benefits and performance of recent offshoring and backshoring decisions; fourth, decision-making and performance related to production relocations; and fifth the company and respondent background.

The questionnaire was jointly developed by researchers in Denmark, Finland, and Sweden. The original survey questionnaire was developed in English and then translated into the local language versions, i.e. Danish, Finnish, and Swedish. Each respondent was given an option to answer either in English or their native local language. The questionnaires were piloted with

test respondents including both researchers not directly involved with the project and practitioners representative of the targeted respondent group. Some items were modified as a result of piloting. For example, as a result of the feedback from practitioners, it was emphasized that the responses concerning offshoring and backshoring decisions concerned the most recent significant and permanent movements of production either abroad or back home. This is an important distinction from the continuous allocation of production among an international plant network, which is a common practice for large international manufacturing companies in certain industries.

This paper focuses on three aspects of the production relocation questionnaire: the extent of offshoring and backshoring; the drivers of offshoring and backshoring; and comparison of backshoring companies with the “only offshoring” and “no-movement” companies in light of company characteristics and activities. Offshoring and backshoring refer to transferring production permanently from one geographic location to another location, either from Finland to another country (offshoring) or bringing it back to Finland (backshoring). Backshoring does not necessarily mean that the same production had been earlier moved abroad. In both offshoring and backshoring the ownership of the transferred production may or may not change. The survey covered production location movements during the preceding five years. The most recent significant relocation instances, both offshore and back, were studied more specifically in cases when the company had implemented these production relocations. Further, both company-level and the focal plant level aspects of offshoring and backshoring were studied. Company refers here either to the entire company or to a specific business area in case of a multi-business corporation.

3.2 Sample and respondents

The targeted companies included all the manufacturing companies with more than 50 employees in all the manufacturing industry categories (ISIC codes 10-33). In Finland, 949 companies belonged to the target group in 2015. All the companies in the target group were contacted by telephone to identify the person responsible for production and operations. The companies' manufacturing contact persons were contacted to ascertain their willingness to participate in the survey. Of the 949 companies 434 were willing to receive the survey, after which an electronic link was sent to the questionnaire. Three reminders were e-mailed to the contact persons to increase the response rate. The number of manufacturing companies finally submitting the completed survey was 242, and of these 13 did not fulfill the criterion of having over 50 employees. Therefore, the final number of acceptable survey responses was 229, corresponding to a 24 percent response rate.

The respondent sample is compared to the total population of manufacturing companies in Tables II-IV in terms of number of employees in the company, industry, and ownership. The respondent sample is somewhat biased toward large companies (over 500 employees) ($\chi^2(3)=86.546$; $p < 0.001$) and one industry is clearly over-represented, namely the machinery industry and equipment (ISIC code 28), even though otherwise the distribution

Distribution across company size	Total population (%)	Respondents (%)
51-100 employees	49.0	31.7
101-250 employees	31.4	31.7
251-500 employees	11.1	11.9
Over 500 employees	8.5	24.7
Total	100.0	100.0

Note: Comparison of the respondent sample to the total industry population of Finnish manufacturing firms in terms of company size (number of employees)

Table II.
Survey respondents in
terms of company size

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Distribution across industry and technology intensity (ISIC industry code)	Total population (%)	Respondents (%)
<i>High technology intensity industries</i>	5.6	7.4
Pharmaceuticals industry (21)	0.8	1.3
Computer, electronic and optical products (26)	4.8	6.1
<i>Medium-high technology intensity industries</i>	32.5	40.2
Chemical industry (20)	5.1	7.0
Electrical equipment (27)	5.5	6.6
Machinery industry and equipment (28)	17.0	22.7
Motor vehicle, trailer and semi-trailer industry (29)	2.6	2.2
Transport equipment industry (30)	2.3	1.7
<i>Medium-low technology intensity</i>	32.5	30.1
Petroleum industry (19)	0.3	0.4
Rubber and plastics industry (22)	6.3	5.7
Other non-metallic mineral products industry (23)	5.0	4.8
Basic metals industry (24)	2.8	1.7
Fabricated metal products, except machinery and equipment (25)	14.0	14.8
Repair and installation of machinery and equipment (33)	4.1	2.6
<i>Low technology intensity</i>	29.4	22.3
Food industry (10)	8.3	6.1
Beverage industry (11)	1.0	0.4
Textile and clothing industry (13, 14)	1.6	1.3
Timber industry (16)	6.4	5.7
Paper industry (17)	4.3	2.6
Graphical industry (18)	3.4	2.6
Furniture industry (31)	2.5	2.2
Other manufacturing (32)	1.9	1.3
<i>Total</i>	100.0	100.0

Table III.
Comparison of the respondent sample to the total industry population of Finnish manufacturing firms in terms of industry

Ownership	Total population (%)	Respondents (%)
Finnish privately owned	75.3	75.1
Foreign owned	23.5	24.9
Finnish state owned	1.2	0.0
Total	100.0	100.0

Note: Comparison of the respondent sample to the total industry population of Finnish manufacturing firms in terms of ownership

Table IV.
Survey respondents in terms of ownership of the company

follows the population distribution ($\chi^2(20)=14.371$; $p=0.811$). The distribution of ownership in the sample matches with that in the population ($\chi^2(1)=0.125$; $p=0.757$).

Of the survey respondents 84 percent were production managers, plant directors or managers, global operations directors or managers, and supply chain directors or managers. Other responsibilities of the respondents included, e.g. chief executive officers, chief financial officers, chief procurement officer, quality and development manager, and chief technology officer. The average work experience of the respondents in production and operations management tasks was 15.5 years and in their respective current job positions 6.5 years. Therefore, we can conclude that the respondents had in-depth experience and knowledge to answer the questions in the survey questionnaire concerning their company's manufacturing network, strategy and performance, and the recent offshoring and backshoring decisions.

3.3 Questionnaire instrument and variables

The first aspect of this paper, the extent of production movements, was approached by asking if companies had permanently offshored and/or backshored their production during the last five years (yes/no alternatives). Further, the main market areas of the latest significant production transfer and the geographical areas to/from which this production was moved were charted. The following classification of geographical areas was utilized: Nordic countries, other areas in Western Europe, Eastern Europe, North America, Latin America, China, India, Asia (excluding China and India) and other areas (Africa, Middle East and Australia). Expectations regarding production movements in the company during the next two years were elicited using a five-point scale (not at all, small extent, moderate extent, large extent, very large extent) with an additional “not applicable” option. The distributions of the survey responses are presented in Section 4.1.

The second aspect of production relocation in this paper included questions about the company strategies and production relocation drivers. Regarding the strategies, the respondents were asked if their company had a corporate-wide relocation strategy guiding offshoring and backshoring decisions, and if strategies for the focal plant and corporate-wide manufacturing had been created (yes/no). Drivers of the latest significant production location movement were elicited by exploring 21 aspects defined as factors contributing to relocation decisions in the earlier research literature (e.g. Kinkel and Maloca, 2009; Stentoft *et al.*, 2015; Tate *et al.*, 2014). These drivers are presented in Figure 2 in Section 4.2. The impact of the drivers on the recent significant production relocation decision was charted for both offshoring and backshoring using a five-point scale (very low, low, neither high nor low, high, very high) with an additional “not applicable” option.

For the third aspect, being a comparison of backshoring companies with other companies, some background aspects of the companies were ascertained. First, company size was measured in terms of the number of employees in the focal plant and at the company level, as well as in terms of the number of own production plants in the company’s manufacturing network. The classifications of 1-50, 51-100, 101-250, 251-500 and over 500 employees were used in the focal plant review; the first of these groups was discarded when studying the company level. The number of plants was determined by a classification of one, two, three to five, six to ten and over ten plants. All these questions had also a “not applicable” alternative. The industries of the responding companies were classified according to the Eurostat (2016) codes into four categories according to technological intensity (high technology, medium-high technology, medium-low technology and low technology). The distribution of the companies according to the number of employees in a company and the industries and their technological intensity are presented in Tables II and III, respectively.

3.4 Data analysis

The data were analyzed in order to study the extent of production movements, drivers of these movements, and characteristic of companies that have backshored production. The extent of production relocations was analyzed through a descriptive analysis of response frequencies (*n* or %). The importance of offshoring and backshoring drivers were reviewed by descriptive statistics and compared using Mann-Whitney *U*-test. The differences in importance of drivers between large and small companies as well as high and low technological-intensity companies was analyzed also by Mann-Whitney *U*-test. For these analyses the company size (number of employees) and technology intensity was classified as binary variables due to the limited number of respondents particularly in the group of backshoring companies (250 employees at maximum and more than 250 employees; low/medium-low technology intensity and high/medium-high technology intensity).

Further, the characteristics of companies that reported having brought their production back to Finland were compared with the groups of “only offshoring” and “no movement” companies.

The focus was on three aspects known to affect production relocations: industry (technology intensity), company size (number of employees in the company and the focal plant; number of production plants), and existence of a strategy guiding production relocations. The objective was to understand if the backshoring companies together formed a group with some specific characteristics. The descriptive results are presented in terms of category frequencies (n or percent) or scale averages. In addition to the descriptive inspection, multinomial logistics regression was used to identify the characteristic of the backshoring companies.

4. Results

4.1 Extent of production relocations

We first examined the extent of offshoring and backshoring to understand the status of location movements among the respondents. The survey results showed that a considerable amount of production had been transferred across national borders in both directions: 78 out of the total 229 companies (34 percent) had realized some production movements during the last six years (2010–2015). However, more production was moving away from Finland than coming back. Out of the total 229 respondent companies 59 (26 percent) reported having permanently moved production to other countries during the last five years and 30 companies (13 percent) reported having moved production to Finland. Out of the 78 companies that had moved production abroad or back, 11 companies (5 percent of all respondent companies) reported having engaged in both offshoring and backshoring during the time period analyzed. In total, 151 companies (66 percent) reported having neither offshored nor backshored production during the past five years.

Most of the selected 59 latest significant offshoring movements were made to Eastern Europe or Western Europe including the Nordic countries. The next typical location of production offshoring was China. The most recent significant backshoring cases of bringing production to Finland were mostly from Western Europe or another Nordic country, Eastern Europe or China. The detailed percentages and comparison between offshoring and backshoring locations are presented in Figure 1.

Out of the 59 companies that had offshored their production during the reviewed period, 32 (54 percent) had made three or more production moves. Only one move was made by 15 (25 percent) of the offshoring companies. The production was moved to both own company's plant outside of Finland (79 percent of the companies) and to another company's plant (54 percent of the companies, some companies had made simultaneous moves to both own and another company's plants). Backshoring was more infrequent, thus only six (20 percent) out of the 30 backshoring companies had moved production back to Finland three or more times. About 40 percent of the companies had moved production back only one time. Production was moved back quite as commonly from own company's plant abroad (67 percent of the companies) than from an external supplier or contract manufacturer (64 percent of the companies).

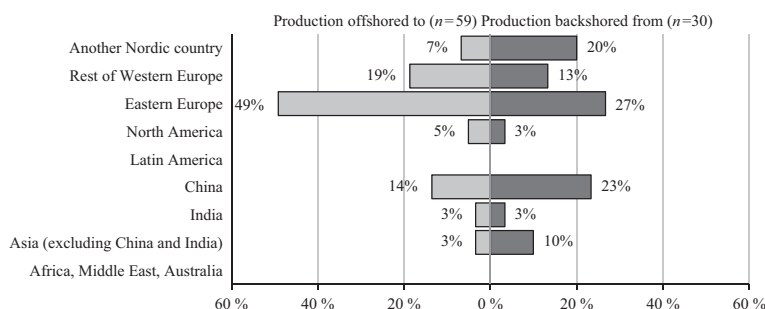


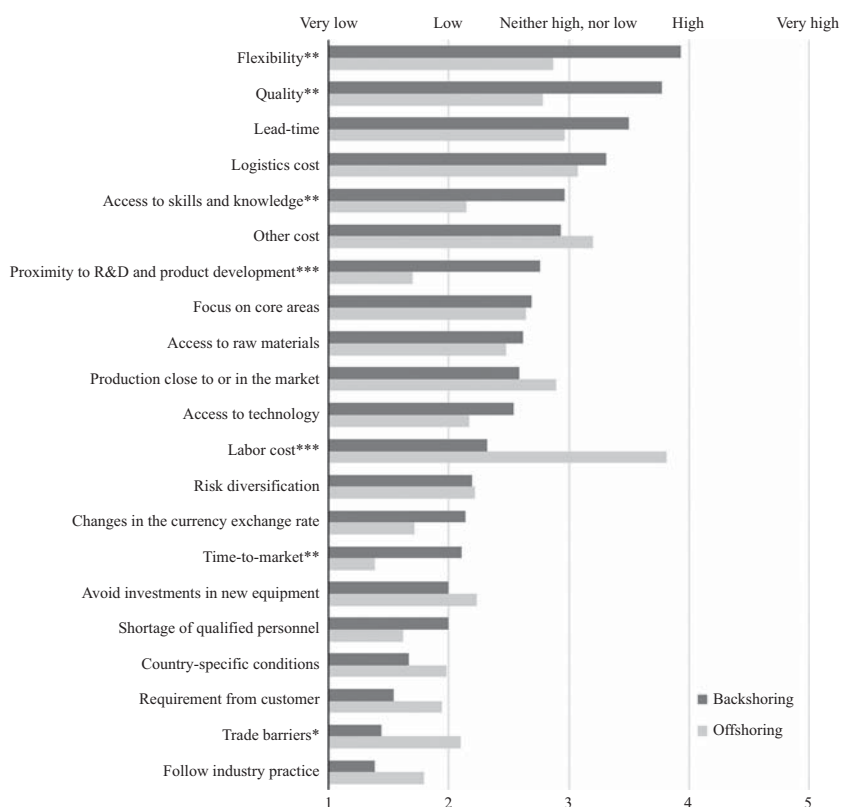
Figure 1.
Comparison of
offshoring and
backshoring
movements regarding
the geographical areas

4.2 Drivers for offshoring and backshoring

According to the survey responses, the most important reasons for offshoring from Finland were labor, logistics, and other (not labor and logistics) costs. The only factors that had a statistically significantly higher rating for offshoring compared to backshoring were labor costs ($U=335.0$; $p<0.001$) and trade barriers ($U=455.0$; $p<0.05$). In turn, flexibility, quality, lead time, and logistics costs were the most important factors when bringing production back to Finland. Out of these flexibility and quality had higher ratings compared with offshoring ($U=434.5$; $p<0.001$ and $U=515.5$; $p<0.01$, respectively). Other drivers rated more significant among backshoring companies than offshoring were proximity to R&D and product development ($U=430.5$; $p<0.001$), access to skills and knowledge ($U=485.5$; $p<0.01$) and time-to-market (bringing new products to market faster) ($U=486.0$; $p<0.01$). Figure 2 shows the reported importance of factors for making the company's latest significant offshoring and backshoring decisions.

We conclude that offshoring decisions of Finnish manufacturing firms are dominated by labor costs, whereas drivers of backshoring decisions are clearly more numerous, including both operative factors (flexibility, quality) and factors related to knowledge and capabilities (access to skills and knowledge, proximity to R&D and product development, and time-to-market).

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Notes: Averages of all responses, $n=59$ offshoring/30 backshoring. *** $p<0.001$;

** $0.001 \leq p < 0.010$, * $0.010 \leq p \leq 0.50$

Figure 2.
Importance of factors
in offshoring and
backshoring decisions

JMTM

4.3 Companies bringing production back to Finland

Technological intensity of industry. The group of 30 companies that reported having brought production back to Finland was analyzed to compare them with the group of only offshoring and no movement companies. Companies having implemented backshoring could be found in 11 out of the 20 ISIC manufacturing industry categories. The highest number of backshoring companies was in the machinery and equipment industry (ten companies, ISIC code 28), fabricated metal products (four companies, code 25), electrical equipment (three, code 27), computer, electronic and optical products (three, code 26), and chemicals industry (three, code 20). The companies that had transferred their production back to Finland were more commonly in industries with higher technology intensity than those companies that had only offshored their production or not realized production movements. The distributions of companies according to their industry's technology intensity in the groups of backshoring, only-offshoring and no-movement companies are compared in Table V.

Company size. Companies in the groups of backshoring, only-offshoring, and no-movement were compared in terms of the number of employees in the company and the number of production plants. Backshoring companies had more commonly a higher number of employees (40 percent had more than 500 employees) than the no-movement companies (22 percent had more than 500 employees) but the highest share of large companies was among only-offshoring companies (49 percent had more than 500 employees). Companies having implemented backshoring to Finland had a higher number of production plants than the only-offshoring and no-movement companies. One-third (ten) of the backshoring companies had over ten production plants, compared to about ten percent of the other companies.

Strategies. In total, 96 companies (42 percent) of all the 229 participating companies reported having a corporate-wide strategy for guiding offshoring and backshoring decisions. As Table V shows, a larger share of the backshoring companies (77 percent) reported having a corporate-wide strategy for guiding offshoring and backshoring decisions than did only-offshoring (65 percent) and no-movement companies (29 percent).

	Backshoring companies (<i>n</i> = 30) (%)	Only-offshoring companies (<i>n</i> = 48) (%)	No-movement companies (<i>n</i> = 151) (%)
<i>Technology intensity of the industry</i>			
High/medium-high	70.0	56.3	40.4
Medium-low/low	30.0	43.8	59.6
<i>Number of employees in the company</i>			
51-100 employees	20.0	14.9	39.3
101-250 employees	26.7	25.5	34.7
251-500 employees	13.3	10.6	12.0
Over 500 employees	40.0	48.9	14.0
<i>Number of production plants</i>			
1 plant	23.3	14.9	45.3
2 plants	10.0	17.0	21.3
3-5 plants	23.3	36.2	20.7
6-10 plants	10.0	19.1	4.7
over 10 plants	33.3	12.8	8.0
Company has a corporate-wide strategy for guiding offshoring and backshoring decisions	76.7	64.6	28.6
Finnish privately owned company	66.7	72.9	77.5

Table V.
Characteristics of the
backshoring
companies vs
only-offshoring and
no-movement
companies

Ownership. The share of Finnish privately owned companies was slightly lower among companies that had moved production back to Finland (67 percent) compared with the companies that had only offshored their production (73 percent) or that had not made any production movements (77 percent) (Table V).

Table VI shows correlations between the variables included in the multinomial regression analysis of the company characteristics on production movements. The ownership variable was omitted from the final regression model, thus inclusion of it did not improve the model. Table VII presents the results of the regression analysis. The results suggest that companies that have not made production movements are more likely from industries with low or medium-low technology intensity when compared with backshoring companies (coeff. = 1.636, $p < 0.010$, OR = 5.133). This relation was not found as statistically significant when comparing the groups of only-offshoring and

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	1	2	3	4
(1) Production movement types				
(2) Technology intensity	0.20**			
(3) Number of employees in the company	0.28***	0.30		
(4) Number of manufacturing plants	0.30***	-0.02	0.589***	
(5) Corporate-wide strategy for guiding offshoring and backshoring decisions	0.38***	-0.00	0.29***	0.29***

Notes: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

Table VI.
Pairwise correlations

	Only-offshoring ($n = 46$)		No-movement ($n = 145$)	
	Coefficient (SE)	OR (95% CI)	Coefficient (SE)	OR (95% CI)
<i>Technology intensity</i>				
Low/medium low	0.75 (0.55)	2.12 (0.73/6.19)	1.64** (0.51)	5.13 (1.90/13.85)
High/medium high	0		0	
<i>Number of employees in the company</i>				
51-100 employees	-1.77 (1.00)	0.17 (0.02/1.21)	-0.49 (0.93)	0.61 (0.10/3.80)
101-250 employees	-1.43 (0.81)	0.24 (0.05/1.18)	-0.27 (0.79)	0.77 (0.16/3.58)
251-500 employees	-1.71 (0.97)	0.18 (0.03/1.22)	-0.48 (0.91)	0.62 (0.10/3.66)
Over 500 employees	0		0	
<i>Number of manufacturing plants</i>				
1 plant	1.87 (1.07)	6.47 (0.80/52.24)	1.94* (0.94)	6.95 (1.09/44.21)
2 plants	2.93** (1.11)	18.79 (2.13/165.98)	2.37* (1.03)	10.68 (1.43/79.83)
3-5 plants	2.29** (0.88)	9.84 (1.75/55.27)	1.09 (0.84)	2.99 (0.58/15.39)
6-10 plants	1.96* (0.91)	7.08 (1.20/41.92)	0.37 (0.95)	1.44 (0.22/9.34)
Over 10 plants	0		0	
Corporate-wide strategy for guiding offshoring and backshoring decisions				
No	0.76 (0.59)	2.14 (0.68/6.76)	2.15*** (0.53)	8.61 (3.04/24.4)
Yes	0		0	
Intercept	-0.86 (0.60)		-1.18* (0.57)	
R^2 Nagelkerke	0.37			
AIC	207.5			
χ^2	81.7***			

Notes: Reference group = Backshoring ($n = 30$). * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

Table VII.
Multinomial logistic
regression on
production
movements

backshoring companies. Further, the company size measured as the number of employees in the company was not statistically significantly related to the decisions of only-offshoring or no-movement groups when they were compared with the backshoring companies. However, significant relationships were found when reviewing the number of manufacturing plants. When compared with the backshoring companies, the companies that had not made production movements more often had one to two plants (coeff. = 1.93/2.37, $p < 0.050$, OR = 6.95/10.68), but among companies that have only offshored production, two to ten plants were more typical (coeff. = 1.96-2.93, $p < 0.010$, OR = 7.08-18.79). In addition, a strategy guiding offshoring and backshoring decisions was found to have a relation with production movements. The companies that had not moved their production had less commonly a strategy guiding their production movements compared with the backshoring companies (coeff. = 2.15, $p < 0.001$, OR = 8.61). The model classified 71.5 percent of the cases correctly (93 percent of the no-movement companies, 28 percent of the only-offshoring companies and 33 percent of the backshoring companies).

5. Discussion

5.1 Extent of production movements

A considerable amount of production has been transferred across national borders of Finland in both directions. The results show that about one-third (34 percent) of the manufacturing companies have been active in transferring production both offshore and back. More production has been moving away from Finland than coming back; for every company that backshored its production there were 2.3 companies offshoring production (companies having done both offshoring and backshoring have been removed from this ratio). This means that backshoring accounts for 43 percent of offshoring. Surveys that have captured similar type of data have been carried out in New Zealand (Canham and Hamilton, 2013), Europe (a mix of European countries; Dachs and Kinkel, 2013), Germany (Kinkel and Maloca, 2009; Kinkel and Zanker, 2013; Kinkel, 2012, 2014), and the USA (Tate *et al.*, 2014).

The ratio between backshoring and offshoring in Finland seems to be higher compared to what has been reported in earlier studies in other countries. Research in Germany shows that the amount of production backshoring to Germany during the period 2006-2012 was 10-35 percent of offshoring activities (Kinkel, 2014). In the mix of European countries studied by Dachs and Kinkel (2013), the ratio between offshoring and backshoring was about 30-35 percent, and in New Zealand the ratio was 16 percent according to the data collected between 2001 and 2011. In the USA, 21 percent of the US manufacturers said that they were moving manufacturing back or planning to do so (Brennan *et al.*, 2015). The numbers between countries are not directly comparable because they have been collected at different times and using different survey tools. However, they can be taken as indicative of the similarities or differences across the countries.

Our findings, thereby, indicate that the patterns of backshoring in a small and open Nordic economy, such as Finland, might differ from those of other countries. The literature review portrayed Nordic manufacturing industries as key sources of economic wealth for the countries, with distant markets and labor systems as challenges to their competitiveness. Where companies have sought for cost advantages in their manufacturing through internationalization and offshoring, also a contrary trend can be found but in a smaller scale. The discovered extent of backshoring in the case of Finland suggests that companies' manufacturing strategies are evolving and priorities might be changing over time.

5.2 Drivers of offshoring and backshoring

The decision-making drivers differed for offshoring and backshoring in several aspects. According to the survey respondents the dominant reason for offshoring from Finland was

labor costs which is well aligned with earlier research (e.g. Kakabadse and Kakabadse, 2002; Kinkel and Maloca, 2009; Kinkel, 2012). In turn, drivers of backshoring decisions are clearly more numerous, including both operative factors such as flexibility and quality (e.g. Bailey and De Propriis, 2014; Canham and Hamilton, 2013; Fratocchi *et al.*, 2014; Gylling *et al.*, 2015; Kinkel, 2012; Moser, 2013; Martinez-Mora and Merino, 2014; Zhai *et al.*, 2016; Wu and Zhang, 2014; Wiesmann *et al.*, 2017) and factors related to knowledge and capabilities, specifically access to skills and knowledge, proximity to R&D and product development, and time-to-market (Arlbjørn and Mikkelsen, 2014; Bailey and De Propriis, 2014; Kinkel, 2014; Pearce, 2014; Tate, 2014; Stentoft *et al.*, 2015).

5.3 Conditions underlying backshoring

Companies having implemented backshoring can be found in all industries. However, the companies that had transferred their production back to Finland were more commonly in industries with high or medium-high technology intensity than those companies that had not backshored production. This finding, combined with the finding that factors related to knowledge and capabilities were important backshoring drivers, signals that utilization of new technologies might be a factor bringing production to a Nordic location such as Finland. This finding still needs to be interpreted with caution but it is in line with the earlier findings by, e.g. Arlbjørn and Mikkelsen (2014), Stentoft *et al.* (2015) and could offer a motive to guide small Nordic countries to invest in technology innovations as a means to drive manufacturing innovations locally.

Backshoring companies in this study were more commonly companies with a higher number of production plants than the only-offshoring and no-movement companies. Contrary to some earlier studies, the size of the company in terms of the number of employees was not found significantly different for companies that had been active in moving production when compared with the no-movers (cf. Kinkel, 2012). Having a high number of production plants suggests that backshoring companies did not necessarily transfer production to close the plant at the location from which production was moved. Instead, this observation suggests that companies assign specific roles to their plants and allocate production to the plants according to these roles (Ferdows, 1997; Brennan *et al.*, 2015).

The backshoring companies reported more commonly having a corporate-wide strategy for guiding offshoring and backshoring decisions than did no-movement companies (cf. Feldmann and Olhager, 2013). This suggests a difference in sustainable decision-making practices between manufacturing firms that are active in moving production vs those that remain passive. Firms that move their production seem to more often have a strategy process guiding their decisions, whereas firms that are more passive do not have explicit manufacturing strategies.

6. Conclusions

6.1 Research contributions

This research makes a contribution to previous research in three ways. First, the backshoring activity in the small and open economies of the Nordic countries, illustrated here by the example of the Finnish manufacturing industry, seems to be higher compared to earlier studies in larger countries. Second, manufacturing companies in the industries with high and medium-high technology intensity are backshoring their production more commonly than companies in the industries with lower technology intensity. The finding suggests that there is potentially an on-going transformation in the manufacturing industry of Finland, with the downsizing of the more labor-intensive and lower technology-intensive industries to be replaced to some extent by the higher technology-intensive industries. Third, manufacturing companies in all size groups can backshore their production, but backshoring activity is higher among the companies with higher number of plants than the activity among the no-movement companies.

6.2 Limitations and proposals for further research

This survey study has limitations that need to be considered when assessing validity of the results. The questionnaire was designed in an international team and the earlier literature was used, likewise well-established measures and scales, to enable comparability with former offshoring and backshoring studies. Significant effort was made to achieve a high response rate and to clean up the data. However, the sample of 229 Finnish manufacturing companies and the small number of offshoring and backshoring cases among the respondents impeded the use of some advanced statistical techniques within the single-country sample. Data are biased toward larger firms (compared to the population). Also, there is over-representation in the data of one industry compared to the population, i.e. machinery industry and equipment (ISIC code 28). These biases limit the generalizability of the findings.

In this study, a company's technology intensity was measured by its sectoral affiliation instead of using a company-level indicator like R&D intensity. It might be that some companies in sectors like machinery and equipment are acting with a high technology intensity and some others in the same sector with a much lower technology intensity. This limitation is acknowledged and needs to be considered in the interpretation of the results.

6.3 Managerial implications and future research

The findings suggest that moving production across national borders is one option of the manufacturing firms to stay internationally competitive. The drivers of offshoring and backshoring differ, and they are different across industries and company characteristics. Companies must prepare their plant location strategies in line with the requirements set by their context and be proactive in preparing strategies for plant relocations. The relevance of several factors, such as cost, quality, flexibility, and access to knowledge and technology, need to be carefully considered when defining location-specific plant roles in the production networks.

Besides the international studies to test the antecedents and consequences of offshoring and backshoring decisions, this study has pointed out the role of manufacturing strategies in connection with location movements. Further research is needed to explain the background factors of potential successes and failures in location choices. Finally, there is need for more comparative studies exploring both offshoring and backshoring in different countries at the same time with the same methodology. This would be the only way to verify if the economic conditions, e.g. the size of the domestic markets, do influence the production relocation activities.

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